

Integrating the Triadic Synthesis Framework into the GRAPHIA SSH Knowledge Graph

A Federated Governance Layer for Community-Validated Linguistic Data

Technical Whitepaper — Proposal for WP2/WP4 Integration

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Version 4 — revised September 2026: corrected against a full audit of the reference implementation

What changed since v1 (August 2026): the proposal below has since been implemented as a reference MVP (public repository, Apache 2.0). Sections 4.4–4.6 are new — they document governance-mechanism refinements (ADR-004/005/006) and community-facing interfaces built after v1 was written. Section 9 ("Implementation Status") is new and reports what has actually been run and tested, distinct from what remains a proposal pending consortium confirmation. What changed in v3 (following an implementation audit): the companion paper's status corrected to "accepted, in press" rather than "published" (its DOI is assigned but not yet confirmed to resolve); the PID scheme in §4.3 now lists all four identifier kinds (g/v/r/s) instead of three; a §4.1 property comment corrected to match the ontology and code exactly; §9 now itemises the full test suite (pytest + Node + Playwright) and the additional verification performed against a real Virtuoso instance, not only the local SPARQL 1.1 stand-in; §4.6 names the static-hosting target (Vercel) explicitly. What changed since v3 (this revision, 16 September 2026): the store facts are corrected — the live public GoTriple Knowledge Graph node (live since 1 September 2026) is served read-only and runs on QLever, while Virtuoso remains the engine named in the D2.2 architecture for the DAP write path; the companion paper is now published (its DOI 10.23726/cij.2026.1829 resolves); and the ADR spelling is uniform (ADR-002). The v3 wording read Virtuoso's role in D2.2 (ADR-002) as a statement about the live public node; v4 separates the architecture's write path from the live read-only node. No claim in v1 or v2 has been withdrawn; several are now more precise because they were checked against the implementation rather than described from memory.

1. Executive Summary

In the D2.2 architecture, the GRAPHIA Data Acquisition Platform (DAP) describes an automated path in which entities and relations extracted from full-text documents by WP4 AI/NLP services are committed directly into the GoTriple Knowledge Graph, with no step for human review, cultural verification, or contributor attribution. This is adequate for high-resource, well-represented content, but it systematically underserves low-resource languages, minority dialects, and culturally specific material — precisely the long tail that the SSH Knowledge Graph is meant to represent.

This document proposes integrating the Triadic Synthesis Framework (Human Contributors + Agentic Scaffolding + DAO-based Governance, described in the companion academic paper, now published (DOI 10.23726/cij.2026.1829) — see the closing reference) as a federated validation layer for this content, connected to GRAPHIA through existing extension points rather than through changes to the authoritative RDF core.

The proposal does not modify the GRAPHIA RDF stores (Virtuoso in the D2.2 architecture; the live public node is read-only on QLever), the TRIPLE Ontology, or any accepted architectural decision (ADR-002). It adds a small set of DAP-facing microservices, one ontology extension namespace, and one external federation node, all reversible and independently governed.

Since v1, this design has been built and tested: a public reference implementation (Apache 2.0) passes 125 pytest tests, plus 18 Node tests and 5 Playwright UI end-to-end tests, and has completed a real end-to-end run — Kafka, a SPARQL 1.1 store, and the governance pipeline together, not simulated separately — confirming the flow in Section 4.2 executes as designed. The SPARQL layer has additionally been checked against a real OpenLink Virtuoso instance — the engine named in the D2.2 architecture for the DAP write path. The live public GoTriple Knowledge Graph node (live since 1 September 2026) is served read-only and runs on QLever (a direction already anticipated by D2.2 §5.2.5's QLever feasibility study and documented in GRAPHIA WP2's QLever Evaluation Report, Zenodo record 20019957), so the write path still requires the

consortium test instance requested in Section 8. Section 9 gives the specifics. The request to the consortium (Section 8) is unchanged in kind, but is now backed by a working system rather than a design on paper.

2. Motivation: The Validation Gap in WP4 Entity Extraction

Section 4.4.5 of the GRAPHIA Technical Architecture (D2.2) describes entity and relation extraction as a fully automated WP4 service: entities are linked to triple:Document instances via schema:mentions and an oa:Annotation body, with oa:creator identifying the extracting service. No mechanism exists for a domain expert or an affected community to confirm, correct, or contest an extracted fact before it is published.

This mirrors what knowledge engineering has long called the input bottleneck (Feigenbaum, 1977): high-quality structured data depends on a narrow pool of expert annotators. GRAPHIA has effectively removed the bottleneck by removing the experts — which trades scale for reliability, particularly for:

- Minority and endangered languages, where generic NLP models trained on high-resource corpora perform poorly;
- Culturally specific entities (local place names, informal titles, oral-tradition references) that general-purpose entity recognisers systematically mis-tag or miss;
- Content where provenance and contributor attribution carry ethical weight — the concern this framework refers to as data alienation.

The Triadic Synthesis Framework was designed to address exactly this gap: it restores a human validation step without reintroducing the centralised editorial bottleneck that made expert-only curation unscalable in the first place.

3. Proposed Integration: A Federated Node, Not a Core Change

GRAPHIA's own architecture is built around federation rather than centralised ingestion (D2.2, Section 2.1): the SSH KG connects autonomous graphs (OpenCitations, EHRI, GESIS, ORKG) through the GRAPHIA Ontology and the SKG-IF interoperability layer, each graph retaining its own governance, schema evolution, and operational rules.

This proposal treats the Federated Epistemic Node (FEN) — the system implementing the Triadic Synthesis Framework — as exactly this kind of participant: an autonomous node with its own DAO governance, reputation system, and validation infrastructure, connected to GRAPHIA only through message-passing and dereferenceable identifiers. No component of FEN's governance stack (DAO, Quadratic Voting, reputation ledger) needs to be hosted, operated, or maintained by GRAPHIA partners.

Three integration surfaces are used, all of which already exist in the D2.2 architecture:

- The Data Acquisition Platform's event bus (Kafka) — a small set of additional consumer/producer processes, following the same asynchronous, non-blocking pattern already used for translation, classification, and annotation services (Section 4.2).
- The TRIPLE Ontology's provenance mechanism — named graphs plus oa:/PROV-style properties (Section 3.4–3.5) — extended with a small, additive namespace rather than modified.
- The Persistent Identifier scheme (Section 4.5) — extended with a dedicated namespace for governance artefacts, following the same ARK + w3id.org pattern already adopted for GoTriple KG resources.

4. Technical Architecture

4.1 Ontology Extension: the gfen: Namespace

A single additive namespace records validation state and governance provenance on existing oa:Annotation instances, without altering triple:Document, triple:Profile, or any core class.

```
@prefix gfen: <https://w3id.org/got/fen/ontology#> .

gfen:ValidationStatus a rdfs:Class .
gfen:pending      a gfen:ValidationStatus .
gfen:validated    a gfen:ValidationStatus .
gfen:disputed     a gfen:ValidationStatus .
gfen:rejected     a gfen:ValidationStatus .

gfen:validationStatus a rdf:Property . # -> gfen:ValidationStatus
gfen:validationMethod a rdf:Property . # -> gfen:QuadraticVoting | gfen:PeerReview
gfen:governanceDecisionId a rdf:Property . # -> dereferenceable PID (see 4.3)
gfen:reputationSnapshot a rdf:Property . # -> dereferenceable PID of the aggregated snapshot
(see 4.3)
gfen:ledgerAnchor      a rdf:Property . # on-chain tx hash, anchor only
gfen:contributorProfile a rdf:Property . # -> triple:Profile
```

Contributor identity is not re-modelled: FEN reuses the existing triple:Profile class, avoiding a duplicate identity layer between GRAPHIA and FEN. The namespace additionally declares an import relationship to GRAPHIA's TRIPLE Ontology; the target IRI is a placeholder pending confirmation with the consortium (Section 8) and does not affect any triple:* class in the meantime.

4.2 Event-Driven Validation Flow

A small set of new DAP-facing microservices — the FEN Bridge (outbound submission and inbound webhook) and the Validation Result Consumer — sit between Entity & Relation Extraction and the Publisher. Publication is never blocked: candidate entities are published immediately with status gfen:pending, exactly as they are today. Validation outcomes arrive asynchronously and update the same named graph in place.

```
doc.normalized.v1
|
v
[ Entity & Relation Extraction ] (existing WP4 service)
| emits candidates, status = pending
v
dap.entities.pending_validation.v1
|
v
[ FEN Bridge ] (new DAP-facing microservice)
| forwards batches to external FEN API
v
[ FEN: Agentic Scaffolding (LLM-guided) -> DAO / Quadratic Voting ]
| governance decision, asynchronous callback
v
fen.governance.decisions.v1
|
v
[ Validation Result Consumer ] (new DAP-facing microservice)
| writes gfen:validationStatus in the named graph
v
dap.entities.validated.v1
|
v
[ Publisher ] (existing service, unchanged)
|
v
Virtuoso (D2.2 write path) / QLever (live public KG, read-only) + GoTriple indexes
```

This preserves the eventual-consistency principle stated in Section 4.1 of D2.2: if a candidate entity is never voted on, it remains published with status `gfen:pending` indefinitely — there is no failure mode in which the pipeline stalls. Section 9 reports a full, real execution of this exact flow against a Kafka broker and a SPARQL 1.1 store.

4.3 Persistent Identifiers for Governance Records

Governance decisions receive their own dereferenceable identifiers, following the ARK + `w3id.org` pattern already defined for the GoTriple KG (D2.2, Section 4.5), under a dedicated Name Assigning Authority Number (NAAN) rather than the GoTriple namespace.

```
ark:{FEN_NAAN}/g00042      # governance decision
ark:{FEN_NAAN}/v00042      # validation record (entity + decision + snapshot)
ark:{FEN_NAAN}/r00042      # reputation snapshot at decision time
ark:{FEN_NAAN}/s00042      # Agentic Scaffolding session that produced the candidate

https://n2t.net/ark:{FEN_NAAN}/g00042
  -> redirects to ->
https://w3id.org/fen/id/decision/g00042
```

The `v`, `r`, and `s` identifiers resolve through the same N2T → `w3id` redirect pattern shown above, under `/validation/`, `/reputation-snapshot/`, and `/session/` respectively. Only an aggregated decision record is published at the `g` PID — quorum, outcome, method, and a link to the reputation snapshot — not individual votes. This keeps the governance record GDPR-compatible: no personal voting data is exposed, and only the on-chain anchor (a hash) is immutable, not the underlying content in the RDF store (Virtuoso in the D2.2 path). This scheme is formalised as ADR-003 (Section 6).

4.4 Governance Mechanism Refinements (new since v1)

Building the reference implementation surfaced three governance-design questions not addressed in v1. Each is resolved as a separate Architecture Decision Record so the core proposal in Sections 1–4.3 remains unaffected by how they are ultimately settled.

- AI role boundary (ADR-004, accepted): an optional LLM-based reviewer may recommend an outcome to speed up triage, but it never votes, never renders a verdict, and never writes `gfen:validationStatus` — the DAO decision is the only source of a final outcome. This directly addresses the risk that a review layer marketed as "community governance" could quietly become automated moderation by another name.
- Participation incentives and small-community fallback (ADR-005, accepted): quadratic voting is only meaningful above a validator-count threshold; below it, the mechanism degenerates into an ordinary committee without the collusion resistance it is meant to provide. ADR-005 sets an explicit threshold, falls back to `gfen:PeerReview` below it, and specifies reputation as a portable, non-tradeable incentive rather than a token, to avoid financial-instrument classification risk in an EU regulatory context.
- Dispute window (ADR-006, DRAFT — not yet accepted): a proposal for a time-bounded, tokenless challenge mechanism using a reputation lock instead of financial stake, so a contested decision can be reopened before it is treated as final. This is included here for transparency about the framework's direction; it is explicitly not part of the integration being requested in Section 8 until it is accepted.

4.5 Data Integrity Validation (new since v1)

The `gfen:` namespace is accompanied by a SHACL (Shapes Constraint Language) shape file that validates the structural well-formedness of governance records — for example, that a `gfen:GovernanceDecision` always references exactly one outcome and one reputation snapshot. This runs as an automated check in continuous integration against representative RDF fixtures, catching malformed governance data before it could reach a

shared graph, independently of whatever store holds the data (Virtuoso in the D2.2 production path, a lighter SPARQL 1.1 store in local development; the live public GoTriple KG node is read-only on QLever).

4.6 Community-Facing Interfaces (new since v1)

Two lightweight, framework-agnostic interfaces were built on top of the same backend described in Section 4.2, intended for demonstration and for community-side interaction rather than as a replacement for GRAPHIA's own front ends:

- A validation-status widget — a small embeddable component that displays the current `gfen:validationStatus` (and, once decided, the dereferenceable decision PID from Section 4.3) for a given entity. It can be dropped into any existing page, including a GRAPHIA-facing one, without requiring changes to that page's own stack.
- A community portal — a minimal interface for submitting candidates, casting votes under the configured method (Section 4.4), and viewing reputation history, used for demonstration and internal testing rather than as a production governance client.

Both are static, zero-build artefacts (no Node toolchain) and can be hosted independently of the backend services; a static-hosting deployment configuration for Vercel is included in the repository. Neither interface is a prerequisite for the Kafka-based integration in Section 4.2 — they are optional, and can be omitted entirely from any consortium-facing pilot.

5. Architecture Decisions

The proposal's boundary is defined by a growing set of Architecture Decision Records (ADRs), following the same lightweight-ADR practice GRAPHIA itself uses (e.g. ADR-002 for the Virtuoso decision). Two anchor the original proposal; three more were added while building the reference implementation. Rows shaded amber are still in draft and are not part of the current request to the consortium.

Decision	Statement	Rationale
ADR-001	Blockchain is used only to anchor a hash of the governance decision. All content remains in the RDF store (Virtuoso as the D2.2 authoritative core; the live public node is read-only on QLever).	Avoids conflict with ADR-002 (Virtuoso as authoritative RDF core) and preserves GDPR right-to-erasure for content, since only a decision hash — not personal or linguistic data — is immutable.
ADR-002	FEN is integrated as an autonomous federation node (Section 2.1 pattern), not embedded inside GoTriple KG or the DAP core.	Matches GRAPHIA's own federation philosophy; requires no GRAPHIA partner to operate or govern DAO infrastructure; keeps the addition fully reversible.
ADR-003	Governance artefacts use the ARK + w3id.org PID scheme described in Section 4.3, under FEN's own NAAN.	Reuses a pattern GRAPHIA has already adopted for GoTriple KG resources (D2.2 §4.5) rather than inventing a new identifier scheme.
ADR-004	An optional LLM reviewer may recommend, but never decide, an outcome (Section 4.4).	Keeps the human/DAO decision authoritative; prevents the framework from becoming automated moderation under a governance label.
ADR-005	Quadratic voting applies only above an explicit validator-count threshold; below it, PeerReview applies. No token economy in the MVP.	Quadratic voting's collusion resistance depends on a minimum participant count; below it the mechanism adds complexity without benefit. Avoiding a token

		sidesteps EU financial-instrument classification risk at this stage.
ADR-006 (DRAFT)	A tokenless, reputation-locked challenge window for contesting a decision before it is final.	Not yet accepted. Included for transparency; excluded from the current consortium request until settled.

6. Risk Assessment and Mitigations

Risk	Description	Mitigation
Sybil attacks	A single actor creates multiple identities to amplify voting power.	Contributor identity reuses triple:Profile; identity verification strength is a FEN-side governance parameter, independent of GRAPHIA.
Governance capture	Early participants accumulate disproportionate reputation-based influence.	Reputation decay and Quadratic Voting cost structure (n^2) cap the influence of any single actor; parameters are tunable per deployment.
Small-community threshold	Quadratic voting degrades into an ordinary committee below a minimum validator count, losing its collusion-resistance benefit.	ADR-005 defines an explicit threshold and falls back to <code>gfen:PeerReview</code> below it, rather than applying quadratic voting universally.
GDPR / immutability tension	On-chain records cannot be deleted once written.	Only decision hashes are anchored on-chain (ADR-001); all deletable content stays in the RDF store (Virtuoso as the D2.2 authoritative core; the live public node is read-only on QLever) under GRAPHIA's existing data policies.
Pipeline stalling	Waiting for community validation could block publication.	Candidates publish immediately as <code>gfen:pending</code> ; validation updates arrive asynchronously (Section 4.2), consistent with D2.2's eventual-consistency model — confirmed by the real run in Section 9.

7. Positioning Relative to GRAPHIA's Own AI Services

GRAPHIA already develops its own LLM-based services for the SSH domain (e.g. an open-weight LLM service, a retrieval-augmented KGQA service, and a human-in-the-loop subject-indexing pilot). This proposal does not duplicate them, and is intended to compose with them rather than compete:

- Where GRAPHIA's own AI services need an underlying model for Agentic Scaffolding (Section 4.4), FEN's scaffolding layer can call them directly rather than operating an independent model fleet — the LLM boundary in ADR-004 applies regardless of which model is behind it.
- GRAPHIA's existing human-in-the-loop patterns are single-expert review; this proposal is collective decision-making (DAO, Quadratic Voting, reputation) — a different governance model aimed at a different failure mode: expert-only review does not scale to the long tail of low-resource languages and culturally specific material described in Section 2, where the pool of qualified single experts is often the limiting factor.
- The niche this proposal targets — community-owned, culturally sensitive data where the affected community is the appropriate authority on correctness — is deliberately narrow, and is not intended as a general-purpose alternative to GRAPHIA's existing curation tools.

8. Request to the Consortium

To move from the tested reference implementation (Section 9) to a pilot integrated with GRAPHIA, the following are requested from GRAPHIA partners: One of them is not a convenience: the live public GoTriple Knowledge Graph node is served read-only on QLever and cannot accept validation writes, so a writable test instance is the only environment in which the integration can be verified end to end.

- A dedicated NAAN (or a sub-range under an existing one) for FEN-issued PIDs, following Section 4.5's registration process.
- Read access to a test-environment Kafka topic mirroring `dap.entities.pending_validation.v1`, scoped to a single low-resource-language WP4 test corpus. Write access to a test Virtuoso-backed store (or to the OKD namespace for the two services) is required alongside it: the live public GoTriple Knowledge Graph node is read-only and runs on QLever (Section 4.2), so it physically cannot accept `gfen: validation writes` — without a writable test instance the core round-trip cannot be verified at all.
- Confirmation that the `gfen: namespace extension` (Section 4.1) does not require GRAPHIA Ontology governance sign-off, since it is additive and does not modify `triple:*` classes; confirmation of the correct IRI for the placeholder ontology import noted in Section 4.1.
- A point of contact within WP4 to align on the exact schema of extracted-entity messages, so the FEN Bridge consumes them without transformation.
- Confirmation of the named-graph URI convention used by the live DAP (Section 3.5 of D2.2), since the reference implementation currently uses a placeholder convention pending this confirmation.

9. Implementation Status

This section reports what has actually been built and executed, kept separate from the proposal above so the two are never conflated. Everything stated here can be independently verified against the public repository referenced in the closing note.

- A reference implementation of every component in Section 4.2 (FEN Bridge, Validation Result Consumer, a stand-in governance service for local testing) is public under the Apache 2.0 licence — the same licence class GRAPHIA's own POSI-aligned governance favours.
- 125 pytest tests pass offline (mocked network and message-bus dependencies, in-memory RDF), plus 18 Node tests and 5 Playwright UI end-to-end tests in CI, together covering the ontology, the PID scheme, the SPARQL update logic, the community-facing interfaces (Section 4.6), and each governance-decision code path in Section 4.4.
- A full end-to-end run has been executed against a real Kafka broker and a real SPARQL 1.1 store together — not each component tested in isolation — confirming that a candidate entity travels the complete path in Section 4.2's diagram and lands as a validated, dereferenceable governance record.
- The SPARQL update logic's dialect compatibility and idempotency (Section 4.2) have additionally been verified as a CI gate against a real OpenLink Virtuoso instance — the engine named in the D2.2 architecture for the DAP write path, not the engine of the live public GoTriple KG node (that node is read-only and runs on QLever) — not only against the lighter SPARQL 1.1 store used for day-to-day local development.
- The SHACL validation described in Section 4.5 runs as an automated gate on every change to the reference implementation.
- The interfaces described in Section 4.6 are implemented and can be demonstrated live on request; no public consortium-facing deployment exists yet, pending the confirmations requested in Section 8.

What remains unverified is exactly what Section 8 asks for: the Virtuoso compatibility check above used a generic instance of the engine the D2.2 architecture names, not GRAPHIA's own deployment (and the live

public GoTriple KG node, read-only on QLever, cannot accept validation writes at all), and none of the above has yet run against GRAPHIA's own Kafka topics or GRAPHIA's own named-graph conventions. The distinction matters and is stated here explicitly rather than left to be inferred.

10. Next Steps

- Pilot: validate the FEN Bridge against a real GRAPHIA test Kafka topic and a single low-resource-language WP4 test corpus, closing the gap named in Section 9.
- Evaluation: compare entity precision/recall before and after community validation on the test corpus.
- Federation: formal registration of FEN as an external node in the SSH KG federation (Section 2.1), alongside OpenCitations, EHRI, GESIS, and ORKG.
- Governance: settle ADR-006 (Section 5) based on pilot feedback before considering it part of any production integration.
- Publication: joint reporting of results as part of WP2/WP4 deliverables, subject to consortium agreement.

Companion document: "Decentralised Agentic Governance: A Methodology for Community-Owned Linguistic Datasets and Knowledge Synthesis" (Riznyk, 2026), published in the CERN IdeaSquare Journal of Experimental Innovation, 10(2); DOI 10.23726/cij.2026.1829. Provides the full theoretical framework underlying this proposal. Reference implementation: public repository, riznykst/fen-triadic-synthesis (Apache 2.0).